Forecasting Hourly Bike Rentals with XGBoost

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Applied ML project | Python · Google Colab | Regression

EXECUTIVE SUMMARY

This project predicts **observed hourly bicycle rentals** in Seoul using time-related and meteorological variables. An XGBoost regressor was evaluated on a chronologically held-out test period. The target is recorded rentals while the system is operating—not total or unmet travel demand.

RAW OBSERVATIONS	MODELING ROWS	INPUT FEATURES	TEST OBSERVATIONS
8,760	8,465	18	1,265

01 Problem definition

Hourly rentals in Seoul vary with the time of day, calendar conditions, and weather, making bicycle allocation and capacity planning more challenging. The prediction target is **Rented Bike Count**; the model is assessed using MAE, RMSE, R^2, and the structure of its prediction errors.

02 Dataset and scope

ITEM	DESCRIPTION
Data source	UCI ML Repository / Seoul Bike Sharing Demand
Observation window	Dec 1, 2017 – Nov 30, 2018 (hourly records)
Raw dataset	8,760 rows · 14 columns (13 candidate inputs + target)
Target variable	Rented Bike Count: number of rentals per hour
Data quality	No missing records or duplicate date–hour entries reported

The records include temperature, humidity, wind speed, visibility, solar radiation, rainfall, seasonal indicators, and holiday information. Only operating-hours observations were used for modeling.

Exploratory patterns in bike rentals

These descriptive figures use all **8,760 original observations**. They show associations, not evidence of causation.

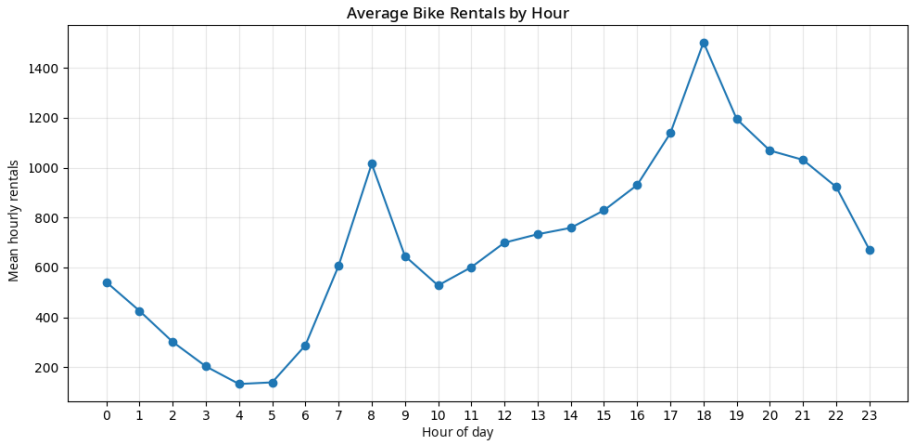


Figure 1. Rental activity has two peaks near 08:00 and 18:00. Average hourly rentals are approximately 1,020 at 08:00 and 1,500 at 18:00.

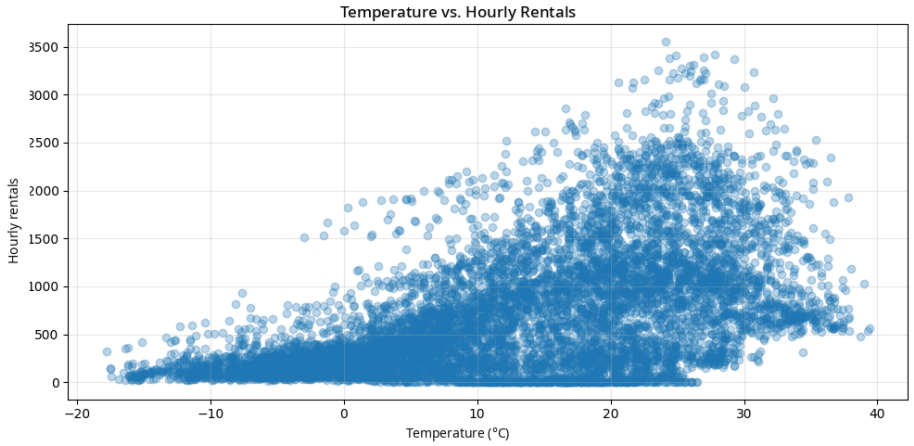
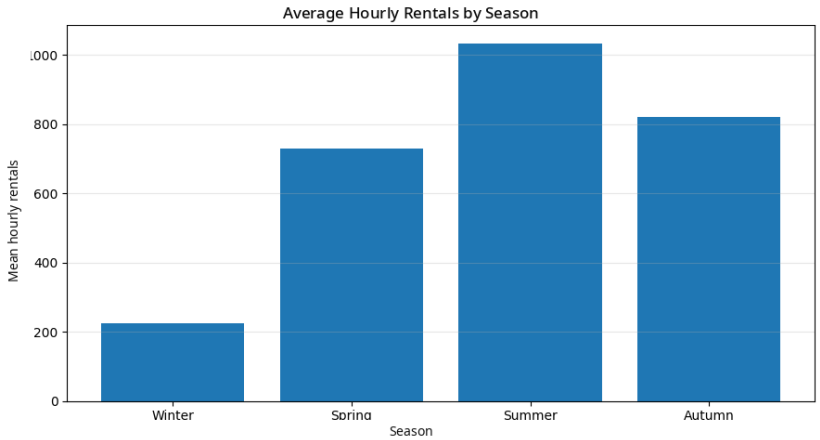


Figure 2. Rentals generally increase as temperature rises, but the association is nonlinear; rental counts vary considerably at the same temperature.



SEASONAL MEANS

Winter	**225.54**
Spring	**730.03**
Summer	**1,034.07**
Autumn	**819.60**

Figure 3. Average rentals per hour are highest in summer and lowest in winter, suggesting that seasonal variables may contribute predictive information.

Preprocessing and modeling approach

03 Data preparation

Filtering: All 295 records from non-operating hours have zero rentals and were excluded to focus on rentals during active service. **Potential outliers:** An IQR rule flagged 152 high observations (about 1.8%); these were retained because they could represent genuine peak periods.

Feature engineering: Month, DayOfWeek, and IsWeekend were derived from the date. Seasons and Holiday were one-hot encoded. The constant Functioning Day column and the raw Date column were excluded. No feature scaling was applied for tree-based XGBoost, leaving **18 numeric predictors**.

04 Chronological train / validation / test split

All hourly observations from the same date were kept in the same subset. This chronological design better reflects predicting a future period using earlier observations.



SPLIT	HOURLY ROWS	DATE RANGE
Train	5,928	Dec 1, 2017 – Aug 6, 2018
Validation	1,272	Aug 7, 2018 – Oct 3, 2018
Test	1,265	Oct 5, 2018 – Nov 30, 2018

Seasonal coverage: Training contains no autumn observations; the test set covers autumn only. The final refit on train + validation incorporates early-autumn data.

05 XGBoost hyperparameter comparison

Objective: **reg:squarederror**. Four candidate settings were ranked by validation RMSE; the test set was not used for model selection.

CANDIDATE	TREES	MAX DEPTH	LR	VAL. RMSE
A · Baseline	300	4	0.10	252.93
C · Lower learning rate	500	4	0.05	260.67
B · Shallower trees	300	3	0.10	269.80
D · Shallow + slower	500	3	0.05	270.01

Final model: test-set performance

Candidate **A** was selected and refitted from scratch on the combined train + validation data (**7,200 rows**). The metrics below use the **1,265-hour held-out test period**, which was not involved in model selection.

TEST MAE	TEST RMSE	TEST R^2
188.83	**274.13**	**0.7531**

MAE indicates about **189 rentals** of absolute error per test hour. The higher RMSE shows that some errors are large. Test R^2 corresponds to a **75.31% reduction in squared error** relative to predicting the test-set mean.

ACTUAL VS. PREDICTED RENTALS

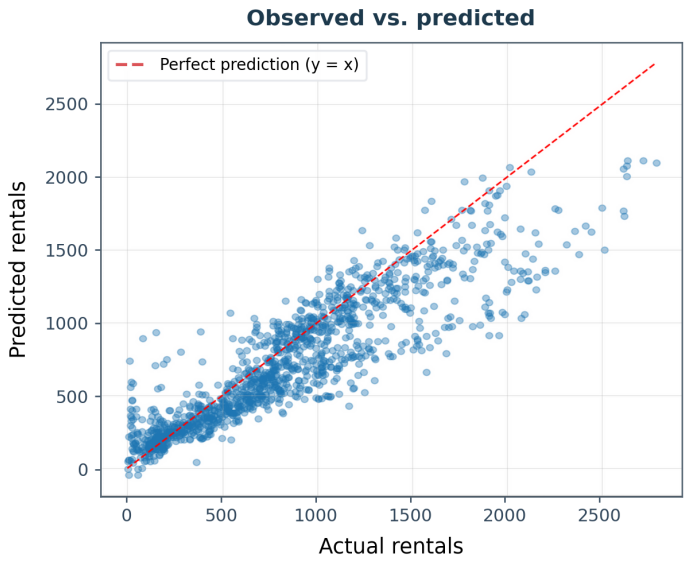
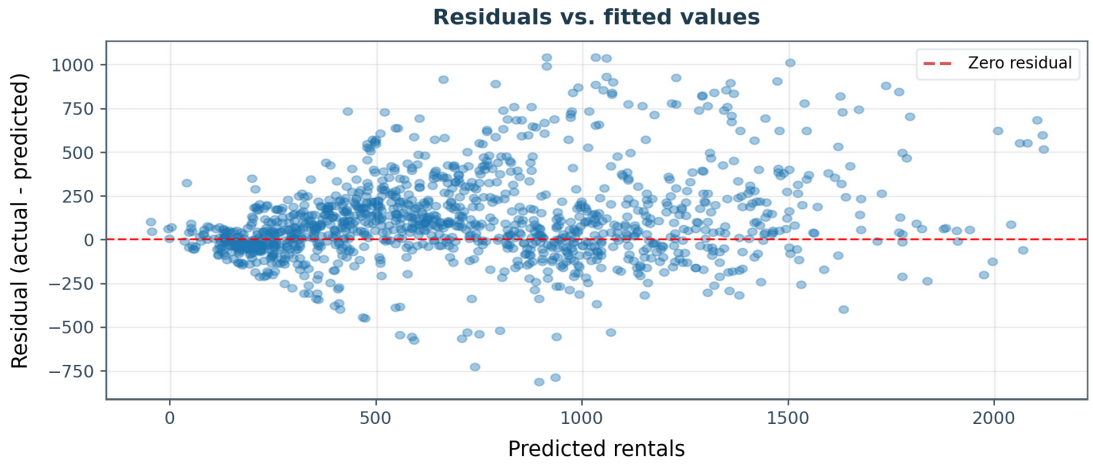


Figure 4. Observed vs. predicted. The diagonal dashed line represents perfect prediction. The model tends to overestimate certain low-rental hours and underestimate the busiest hours. Predictions are generally closer at moderate volumes.

Generalizability: All test observations are from autumn. The reported R^2 should not be assumed to hold for other seasons or years.

RESIDUAL ANALYSIS

Residual = actual - predicted. Mean residual: **+114.43 rentals** (overall underprediction); standard deviation: **249.11**; maximum absolute error: **1,044.29**.



Error analysis and conclusions

MEAN ABSOLUTE ERROR BY HOUR

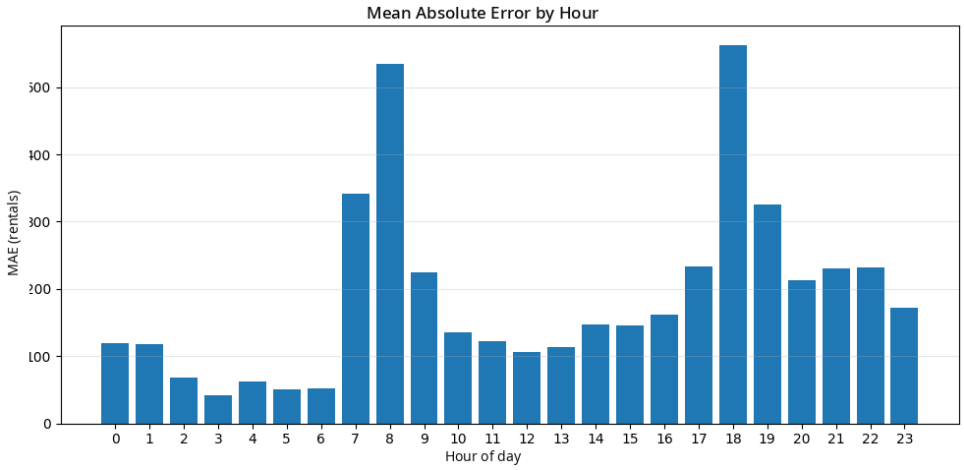
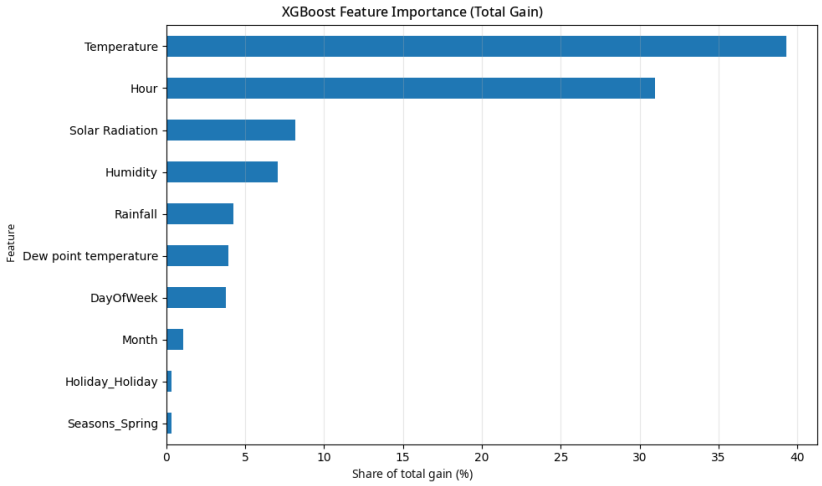


Figure 5. The highest hourly MAE occurs at 18:00 (**562.82**; mean residual **+542.39**) and 08:00 (**533.77**; mean residual **+457.81**). The model substantially underpredicts during the busiest hours.

FEATURE IMPORTANCE · TOTAL GAIN



TOP CONTRIBUTING FEATURES

Temperature	**39.29%**
Hour	**30.95%**
Solar radiation	**8.18%**
Humidity	**7.09%**
Rainfall	**4.27%**

Figure 6. Importance is the share of model total gain, not a causal effect. Temperature and hour jointly account for **70.24%** of total gain.

06 Conclusion and future directions

XGBoost provides a useful baseline for predicting **observed hourly bike rentals** (**test R^2 = 0.7531**). Key limitations are pronounced underprediction during commute peaks, a single-year dataset, and an autumn-only test window. Future work could test multi-year data, rolling-origin evaluation, peak-hour-specific models, and weather predictors that would truly be available at prediction time.

PROJECT NOTEBOOK

Open the project notebook in Google Colab

Dataset: UCI ML Repository (2020) · Seoul Bike Sharing Demand · DOI 10.24432/C5F62R · CC BY 4.0